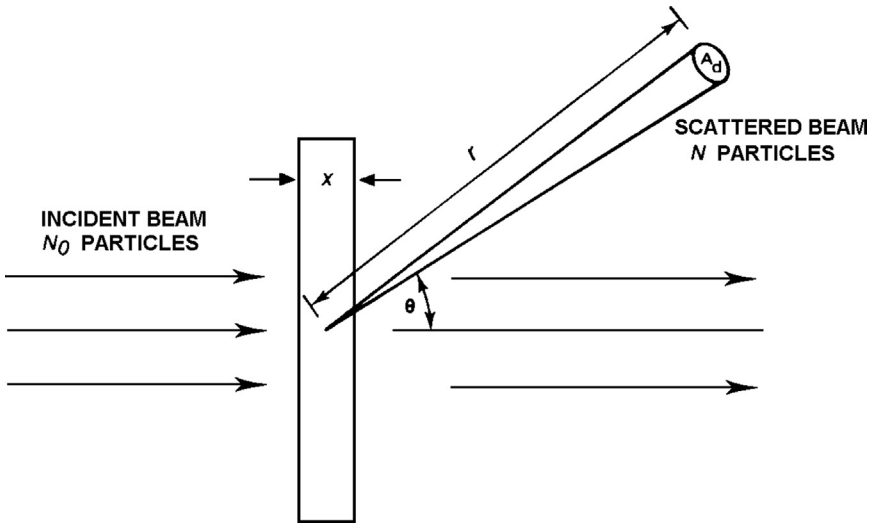


Figure 10.4 Geometry for measuring the scattering by a thin foil.

where n is the number of projectiles scattered into the detector which subtends a solid angle

$$\Delta\Omega = A_{\mathrm{d}}\, r^{-2} \tag{10.20}$$

with respect to the center of the target; A_{d} is the detector area and r is the distance between target and detector; n_{o} is the number of projectiles hitting the target, which is x thick and contains N_{v} scattering atoms per unit volume ($N_{\mathrm{v}} = \rho \mathbf{N}_{\mathrm{A}}/M$, where ρ is the density).

By bombarding foils made of different metals with α-particles from radioactive elements, (10.18) can be used to prove that Z_2 is identical to the atomic number of the metal. This implies that the atom has a central core with a positive charge of that magnitude. Such experiments showed that some particles scattered almost directly back (i.e. $\theta > 90°$), a fact which baffled Rutherford: "It was quite the most incredible event that ever happened to me in my life. It was almost as incredible as if you fired a fifteen inch shell at a piece or tissue paper and it came back and hit you." From the scattering equations he concluded that the scattering center, the "nucleus", had a diameter which was 1000 times smaller than the atom. This same scattering technique was used 55 years later in the US lunar explorer Surveyor 5 (1967) to determine the composition of the lunar surface for the first time.

10.5. ELASTIC SCATTERING

In elastic scattering energy is exchanged between the projectile and the target nucleus but the value of Q is zero:

$$\mathrm{X}_1(v_1) + \mathrm{X}_2(v_2 \approx 0) \;\rightarrow\; \mathrm{X}_1\left(v_1'\right) + \mathrm{X}_2\left(v_2' > 0\right) \quad Q = 0 \tag{10.21}$$

From the conservation of momentum and energy one finds that for a central collision:

$$v_1' = v_1(M_1 - M_2)/(M_1 + M_2) \tag{10.22}$$

$$v_2' = 2v_1 M_1/(M_1 + M_2) \tag{10.23}$$

v_1 is the projectile velocity, and v_1' and v_2' are the velocities of the scattered species.

An important example of elastic scattering reaction involves the slowing down of neutrons in nuclear reactors from high kinetic energies, which they possess when emitted in nuclear fission, to very low energies at which they have much higher reaction probabilities with the nuclear fuel. The neutrons are slowed to energies comparable to those of a neutron gas at the temperature of the material in which they are moving and, hence, they are known as *thermal neutrons*. The most probable kinetic energy for particles at thermal equilibrium at the temperature T is given by the relation $E_{\text{kin}} = \mathbf{k}T$. At 25°C the most probable energy for thermal neutrons is 0.026 eV, while the average energy is approximately 0.040 eV. In nuclear reactions it is necessary to consider the most probable kinetic energy rather than the average energy (cf. §3.6.2).

Equation (10.22) shows that for the case of head-on (or central) collisions between energetic neutrons and protons at thermal energies ($v_1 \approx 0$), the velocity of the neutron is reduced in a single step to thermal energy as a result of the approximately equal masses of the neutron and proton. This effect can be seen in a head-on collision between equally heavy billiard balls. If the target atom is ^{12}C, the velocity of a neutron decreases approximately 15% in a single collision, while the energy of the neutron decreases 28%. If the target atoms are as heavy as uranium, the decrease in velocity and in kinetic energy is on the order of about 1% per collision. Fast neutrons lose roughly their energy in a similar way. However, for ^{235}U fast neutrons lose most of their energy by elastic scattering. (This is important for fast reactors)

The slowing down of energetic neutrons to low kinetic energies is called *moderation*. Equation (10.22) shows that the lightest atoms are the best *moderators* for fast neutrons — (M_1−M_2) smallest — and heavier atoms are poorer moderators because that difference is now rather large. Conversely, heavier atoms will be better *reflectors* for energetic neutrons since they will scatter them back with little loss in energy. For thermal neutrons, however, heavy atoms are not good reflectors because of their greater tendency to absorb the neutrons. Noncentral collisions ($\theta > 0$) are more common than central ones, so the average decrease in neutron energy per collision is less than the values calculated for central collisions (see further Ch. 19).

In a point neutron source, surrounded by a moderator, the thermal neutron flux (particles per unit area and unit time) as a function of the distance from the source is found initially to increase and then to decrease; see Fig. 10.5. The distance from the source at which the flux is at maximum is given in Table 10.1. This is the optimum position for thermal neutron induced reactions.

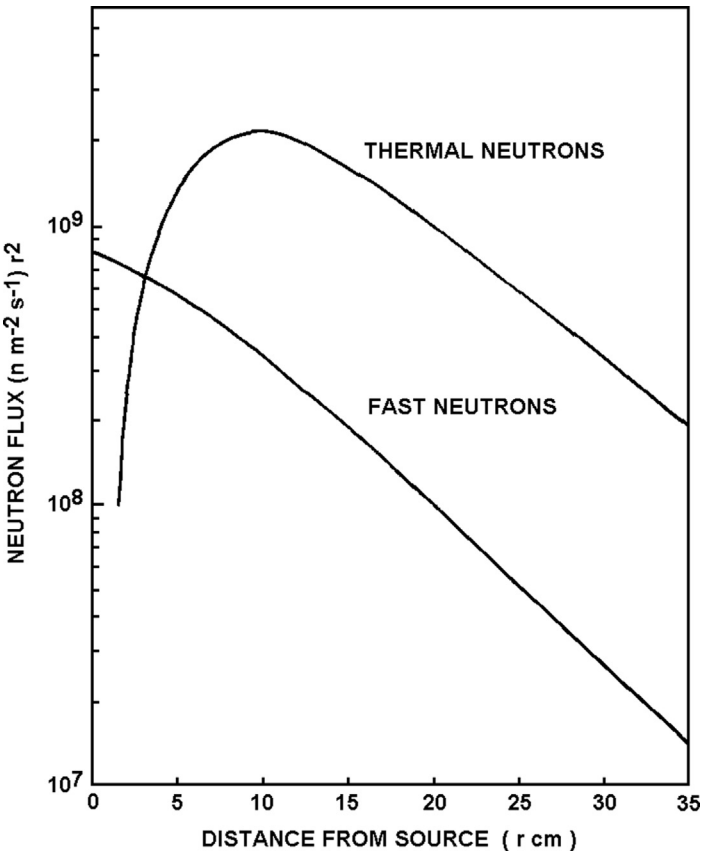


Figure 10.5 The thermalization in water of fast neutrons from a RaBe point neutron source (intensity 10^6 n s ^{1}).

Table 10.1 Distance from neutron source for maximum thermal flux

Neutron source	Neutron energy (at source)	Moderator	
		H_2O	D_2O
RaBe	Average 4 MeV	10 cm	
TD reaction	14 MeV	15 cm	
Fission	Average 2 MeV	7 cm	21 cm

10.6. INELASTIC SCATTERING

If energy is transferred not only to change the velocity of the projectile and the target but also the energy states, excitation, it is called inelastic scattering. Here a part of the kinetic energy of the projectile is transferred to the target nucleus as excitation energy without changing the values of A or Z of either target or projectile. In case the projectile is a